Multi-Distance Phase Retrieval for Polarization-Sensitive Metalens

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***Abstract*—Metalenses offer unique benefits over refractive lenses, such as CMOS compatibility and a high degree of freedom in the design. However, a fast and non-complex method for phase characterization of metalenses, especially for polarization-sensitive or high-NA metalenses, remains a challenge. This paper investigates the Multi-Distance Phase Retrieval method for this purpose. First, a method for designing and simulating polarization-sensitive metalenses is proposed using ANSYS Lumerical. A polarization-metalens was designed with a radius of $100\mu m$, a focal length of $1300\mu m$, and $1500\mu m$ for the different polarizations along the propagation axes, and was designed to work at a wavelength of $800nm$. The Multi-Distance Phase Retrieval was tested with the simulated data from the designed metalens, resulting in an average absolute phase error of $6.6°$ and $6.0°$ for the different polarizations. Also, an extra constraint for the total measurement distance was tested and proven to have a considerable impact on the accuracy of the algorithm. Finally, an experiment was done with the designed lens to obtain measurement data. While the algorithm showed some correlation with the expected phase profiles, it did not produce the desired outputs. This could be caused by high noise, overexposed pixels, and too little information for the algorithm to converge. Experimental data from a different setup produce results close to the expected phase profile. The results show that Multi-Distance Phase Retrieval is a viable method for the phase characterization of polarization-sensitive metalenses, but it is sensitive to noise and needs enough information to converge to the correct phase profile.**

I. Introduction

The potential of metalenses has been seen more and more in the last decade. They have novel properties compared with regular refractive and diffractive lenses that provide significant benefits in specific use cases [1]. Regular refractive lenses are made out of a curved piece of material, which is bulky. Metalenses are composed of nano-pillars, called meta-atoms, that are all the same height, which makes it possible to make them even thinner (around $1\mu m$). The thinness of the metalens is compromised by the thickness of the substrate the lens is built on, which can be around $1mm$. One of the main advantages of metalenses compared with refractive lenses is their compatibility with CMOS technology, making mass production possible, as well as integrating lenses directly onto chips. This reduces coupling losses and makes it possible to make complex yet compact systems [1], [2].

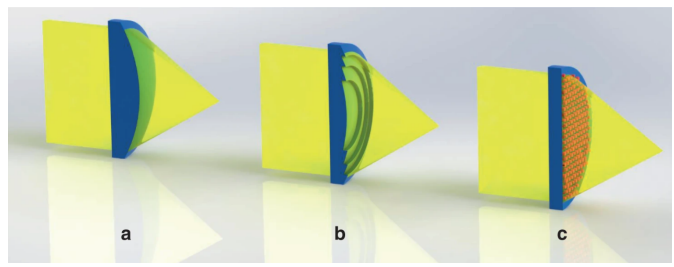


Fig. 1: Difference between refractive lens (a), diffractive lens (b), and metalens (c) from [3].

Diffractive lenses, which are built like a wedding cake, can already be significantly thinner than refractive lenses (also around $1\mu m$). They can also be compatible with CMOS technology, which gives them the same advantages as metalenses. However, metalenses have other unique benefits over diffractive lenses, which make them interesting for complex systems [3]. Because metalenses are composed of individual meta-atoms, they offer greater design freedom for the phase profile than diffractive lenses. This makes metalenses a good choice for arbitrary phase profile design or High numerical apertures (NA). Furthermore, diffractive lenses have limited polarization control. Metalenses, on the other hand, can manipulate the polarization of the light and also be polarization selective, which acts as a filter for the polarization or gives different polarizations a different phase profile [4], [5], [6]. These polarization-sensitive characteristics allow the use of one metalens where there are three or more components needed with traditional lenses. Lastly, some mechanisms can make metalenses tunable, which are not available in diffractive lenses [7].

With the ability of mass production using CMOS processes, there is a growing demand for the characterization of metalenses. The phase profile gives key insight into the working of the lens. It can give information about the focal length, focal power, pupil function, aberrations, and more [8]. Therefore, a fast, repeatable, and noise-robust phase characterization method is needed, which works well for both regular and polarization-sensitive metalenses.

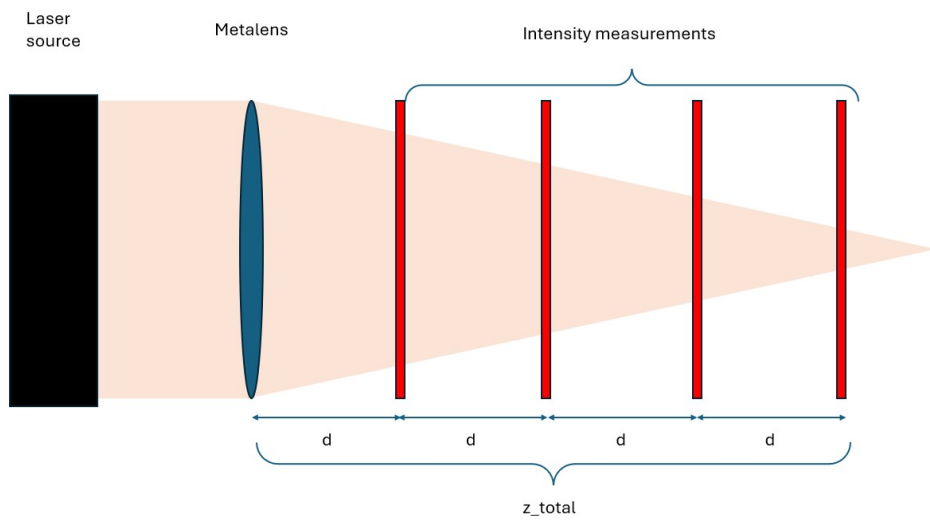


Fig. 2: Multi-Distance Phase Retrieval simplified example setup.

A. Different Phase Characterization Methods

Because of the high frequencies of light, detectors are too slow to directly capture the phase of the light. Phase characterization for light is based on intensity measurement, from which the phase can be extracted using equations or iterative algorithms. There are several ways to characterize the phase of a metalens. The paper by N. Kossowski et al [9] gives a good summary of different methods and their shortcomings. An important fact that is not considered by this paper is that a metalens can have a high NAs, or phase profiles that have abrupt changes. These conditions are not suitable for all of the previously mentioned phase retrieval methods. For example, the transport of intensity equation assumes the paraxial approximation for the beam propagation, which is broken by high-NA lenses [10]. Other methods, like Shearing Interferometry and Shack-Hartmann wavefront sensors, have a limited maximum measurable wavefront slope. If the wavefront slope is too steep, like with a high-NA lens or an abrupt phase change in a metasurface, the spots after the Shack-Hartmann wavefront sensor will overlap with other spots. For shearing interferometry, reducing the shearing distance to capture high-frequency phase variations makes low-frequency components harder to recover. This makes these methods impractical for phase characterization of high-NA metalenses.

Methods that do not require slow-varying phase profiles have other shortcomings. Reference beam interferometry requires a complex setup, a high coherence of the source, and is sensitive to perturbations caused by vibrations. Spatial Ptychography has long measurement times and a complex setup.

Another interesting method is the Gerchberg-Saxton method [11]. This method is based on taking an intensity measurement at the sample plane, which is the plane of the lens, and at the Fourier plane of the lens. The algorithm starts with a random phase profile and the measured amplitude profile, and propagates it to the Fourier plane by taking the Fourier transform of the field. Here, it replaces the calculated amplitude with the measured amplitude in the Fourier plane and propagates it back to the sample plane by taking the inverse Fourier transform. This method is simple and can handle high-NA metalenses as a sample. The downside is that the phase retrieval problem is non-convex, meaning that there are local minima in which the algorithm can get stuck. This makes it possible to converge to the wrong phase profile.

II. Multi-Distance Phase Retrieval

Recently, a phase characterization method has been proposed that offers a solution to these shortcomings [12]. The method claims that it is a simple setup, has fast measurement and calculation times, does not require high temporal coherence from the source [12], can handle high-NA metalenses, and mitigates the non-convexity of the phase retrieval problem by using multiple measurements and averaging over them. This allows the algorithm to overcome local minima and provides improved convergence.

The method is similar to the Gerchberg-Saxton method, only now with multiple intensity measurements at different distances. A light source illuminates the metalens, and a CCD camera is used to collect the intensity at the sample plane and at intervals of d on the propagation axis. These measurements are then used in the algorithm to converge to the correct phase profile. An example of the working principle can be seen in Fig. 2.

The basics of the algorithm are as follows. First, a random phase profile is made and combined with the square root of the intensity measured at the sample plane to get an electric field. This electric field is Fourier transformed to decompose it into plane waves. These plane waves can be individually propagated in the Fourier plane by multiplying them by

$$e^{ik_z z_n} \tag{1}$$

where k_z is the propagation constant, which depends on the plane wave, and z_n is the distance from the sample plane of the n_{th} intensity measurement.

After propagation, the calculated amplitude of the electric field is replaced with the square root of the measured intensity at distance z_n, and the calculated phase is kept. This is done for each intensity measurement. Each electric field, consisting of the propagated phase and measured intensity, is then propagated back to the sample plane, where they are averaged. The phase of this average electric field is kept, and the amplitude is again replaced with the square root of the measured intensity at the sample plane. This is repeated until the difference between iterations is so small that it is determined that the algorithm has converged.

A. Algorithm Functions

The basic working principle of the algorithm is explained above. However, some extra features are added. These either reject erroneous measurements, provide noise robustness, improve convergence speed, or help to check the accuracy of the algorithm.

1) Masking Function: The first function is a masking function that does two checks for the measured intensity per pixel. First, it checks if the measured intensity at a pixel does not exceed the maximum intensity that the CCD pixel can measure. This implies that the pixel is overexposed, losing the information about the intensity. This can happen, for example, if the measurement distance is close to the focal length. For these pixels, the calculated amplitude of the electric field is not replaced by the square root of the measured intensity.

The second part checks if the calculated amplitude of the electric field is within the noise range of the CCD camera. If this is the case, it is undesirable to replace the calculated amplitude with the square root of the measured intensity. The inherent noise of the camera can cause the measured intensity to be a little shifted from the correct intensity. If the algorithm converges to the correct value, it does not take into account this noise. This makes it inefficient to replace the calculated amplitude of the electric field produced by the algorithm with the measured amplitude while it is within the noise range of the measured amplitude.

2) Double Feedback: This function takes into account the result of the two previous iterations to prevent oscillations caused by noise during the measurements. It modifies the back-propagated fields from each measurement to the following formula,

$$g_{s,k} = (1 + a + b)g_{s,k}^{'} - a \cdot g_{s,k-1} - b \cdot g_{s,k-2}, \qquad (2)$$

where $g_{s,k}$ is the back-propagated field after the double feedback, $g_{s,k}^{'}$ is the back-propagated field that was originally produced by the algorithm this iteration, $g_{s,k-1}$ and $g_{s,k-2}$ are the back-propagated fields from the last two iterations, and a and b are parameters that can be set depending on the noise level during the measurements. This is done for each measurement, after which an average of all the fields is taken. This function acts as a dampening function on the oscillations, making sure that the electric field cannot change too fast in one iteration. The best results were obtained with the values $a = 0.6$ and $b = 0.4$.

This interpretation of the double feedback is taken from [13] and has produced good results in this paper. The interpretation of [12], where the double feedback is implemented over the average electric field instead of each backpropagated field individually, was first implemented, but did not produce the desired effect.

3) Kalman Filtering: Kalman filtering is yet another way to make the algorithm more robust against noise. Because of noise during the measurements, the measurements are not the exact correct intensities. After enough iterations, the calculated amplitude of the electric field will become closer and closer to the correct value, which does not include the noise of the measurements. It is therefore advantageous to keep more of the calculated field, the more iterations that have passed. The equation for the Kalman filter is given by

$$E_{s,k} = \left(\omega\sqrt{I_0} + (1-\omega)|E_{s,k}|\right) e^{i\phi_s} \qquad (3)$$

where $\omega\sqrt{I_0}$ is the square root of the measured intensity at the sample plane, $|E_{s,k}|$ the calculated amplitude of the electric field at the sample plane, $e^{i\phi_s}$ the calculated phase at the sample plane, and ω the Kalman variable, which starts at one and linearly goes to zero, where it should be zero at the last iteration.

4) High-noise Measurements: Some measurements may experience above-average noise. This can be caused by vibrations in the measurement platform or the introduction of ambient light. These measurements hinder convergence, and these measurements should be excluded.

This feature first runs the algorithm to its completion without alteration. This will produce a phase profile, which is then used to propagate all of the measurement intensities back to the sample plane. When a measurement experiences above-average noise, it will produce a different electric field than the other measurements. Checking the difference between all of the fields will reveal these measurements, see Fig. 3.

5) Phase Error Check: To measure the accuracy of the algorithm against the expected phase profile of designed lenses, a phase error check was added. The phase error check compares the final phase profile produced by the algorithm against the phase profile that the simulation of the lens produced. If there is no simulated phase profile, it can also produce a perfect phase profile based on the dimensions of the lens and the focal length, and compare the algorithm against this.

The phase error checker lines up the phases by searching for the offset that produces the lowest error. Finally, the phase difference is wrapped to provide accurate results. This method is a little crude compared with, for example, a Zernike expansion. However, because it is expected that the phase profile will be close to the designed phase profile, this will produce accurate results. For completely wrong phase profiles, this is easy to see from visual inspection of the algorithm's

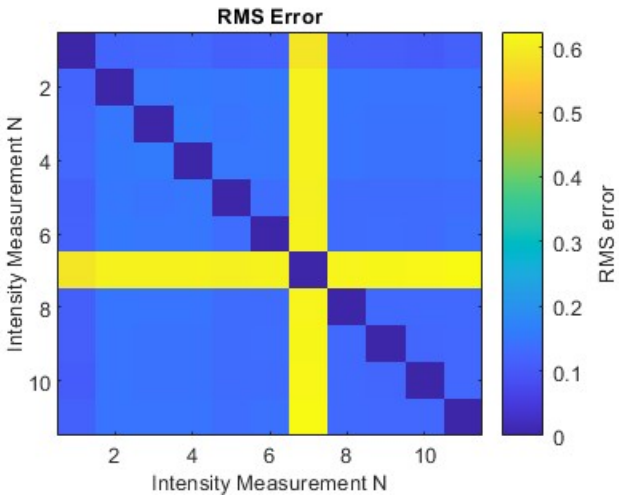


Fig. 3: Example of high-noise measurement checker.

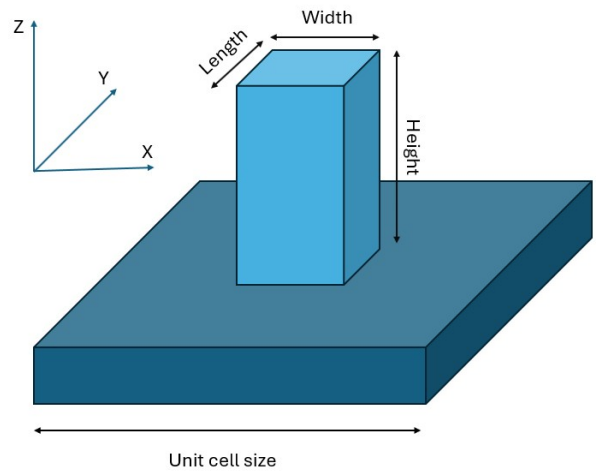


Fig. 4: Unit cell with rectangular meta-atom.

results, and these profiles will exhibit little correlation with the designed phase profile.

III. Metalens Design and Simulation

For this paper, a polarization-sensitive metalens was designed. As discussed in Chapter I, polarization sensitivity is a key advantage of metalenses, and allows for complex systems to be built with one lens. Therefore, it is of importance to test the viability of MDPR as a method for the phase characterization of these lenses.

The metalens design was done with ANSYS Lumerical. RCWA was used to simulate the singular pillars for the lookup table, and FDTD was used to simulate the lens. A detailed guide is available on the ANSYS optics website, which explained how to simulate all the stages of a metalens and how to do the GDS export. This guide is for non-polarization-sensitive metalenses, which use circular pillars where only the radius is varied. Because the independent varying of two different dimensions is needed to create polarization sensitivity, the provided codes had to be rewritten to accommodate these two-dimensional variations. Many of the functions used in the original code were not suited for two-dimensional data, so workarounds needed to be found, some of which were done in MATLAB. The original guide can be found in [14].

A. Polarization-Sensitive Metalens

The polarization of light specifies in what plane the electric field of the light oscillates with respect to the propagation direction. When light is at normal incidence with a lens, the different linear polarizations specify whether the electric field will oscillate parallel to the x-axis or the y-axis of the coordinate system in which the lens is designed. For the rest of this paper, it is assumed that the S-polarization will oscillate parallel to the x-axis and the P-polarization will oscillate parallel to the y-axis.

The most straightforward method of designing a metalens is to use circular pillars and vary the radius of the pillars to achieve different effective indices, which leads to different phase delays. This type of lens looks the same for each polarization, therefore making it insensitive to polarization. By being able to change the x and y dimensions of the meta-atoms independently of each other, the pillars can induce different effective indices for the two different polarizations of the light. Much research is done in this field [4], [5], [6], and the simplest way to achieve this is by using rectangular meta-atoms. An example is given in Fig. 4

B. Design Requirements

There are several constraints for the design of the metalens, which are caused by the physical requirements for the metalens to work or the fabrication requirements. For the fabrication, it was preferred that the meta-atoms would be made from TiO2, since previous lenses had been successfully designed with this material, and the ratio between the smallest and largest dimensions should be less than or equal to 1:10. This constraint is caused by the difficulty of etching such a densely packed pattern. The maximum height was $1\mu m$. Lastly, the measurement setup has a laser that is near infrared. For this reason, the lens will be designed for a wavelength of $800nm$.

To ensure the correct operation of the metalens, constraints on the unit-cell size are imposed. The unit cell is the square of the substrate, which houses one pillar, as illustrated in Fig. 4. The size will determine how close the meta-atoms are spaced. The two requirements for the unit cell size are given in [15]. To ensure only zero-order diffraction,

$$U < \frac{\lambda}{n_{substrate}}, \tag{4}$$

where U is the size of the unit cell, λ the wavelength of the light, and $n_{substrate}$ the refractive index of the substrate. If this requirement is not met, the higher-order diffractions will diffract the light at multiple larger angles than intended. The second requirement, which is based on the Nyquist criteria, is as follows,

$$U < \frac{\lambda}{2\mathbf{NA}}, \tag{5}$$

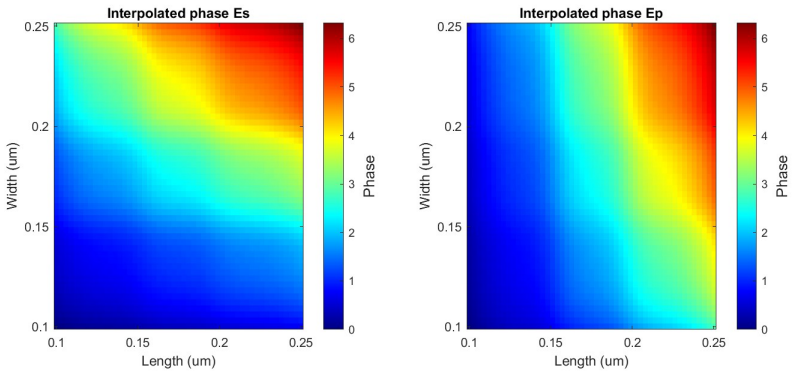


Fig. 5: Lookup table for the different polarizations.

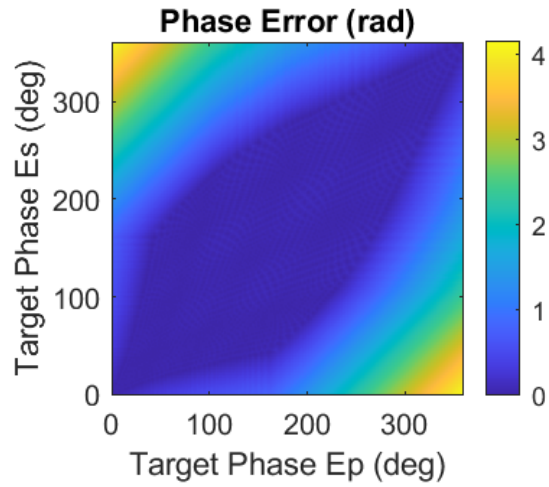


Fig. 6: Phase error for combinations of the phases in the different polarizations.

where **NA** is the numerical aperture of the metalens. The strictest requirement is set by (4), unless the NA exceeds $n_{substrate}/2$. With the wavelength being $800nm$ and the refractive index of the substrate being 1.453, this leads to a maximum unit cell size of $0.55\mu m$.

C. Lookup Table

The metalens will be comprised of meta-atoms with varying width and length, to impart different effective indices and therefore different phase shifts for the different polarizations. To obtain the phase shift corresponding to the dimensions of the meta-atoms, a lookup table is built. This is done in ANSYS Lumerical RWCA. First, a general model of the rectangular pillar is made, and the simulation is run. Next, a parameter sweep is done for different lengths and widths, while keeping the height constant. The minimum and maximum length and width will be determined by the fabrication ratio of 1:10. The minimum will be

$$W_{min} = L_{min} = \frac{H}{10}, \tag{6}$$

where W is the width, L is the length, and H is the height of the meta-atom. The maximum is determined by the spacing that should be left between neighbouring meta-atoms, resulting in

$$W_{max} = L_{max} = U - \frac{H}{10}, \tag{7}$$

where U is the unit cell size. The imparted phase is then extracted and exported to MATLAB for further processing.

The extracted phase is unwrapped using the MATLAB script and normalized so that the smallest meta-atom will impart zero phase. To keep the simulation time manageable in the RCWA simulation, the number of steps between the maximum and minimum lengths and widths is set to 11, resulting in $11^2 = 121$ simulations. This results in a low resolution for the phases of the meta-atoms. For this reason, the phase is interpolated with extra points, resulting in a larger lookup table. The final lookup tables can be seen in Fig. 5.

Last, the data is flattened for faster searching in the table, and a lookup function is created. The lookup table takes the phase for the two polarizations as the input, and gives the resulting length and width of the meta-atom needed for these phases. The function checks for the lowest error in the two flattened phase tables and gives the corresponding dimensions.

Along with the phase, the electric and magnetic field of the meta-atoms is also extracted and interpolated. This will be used in the near-field stitching simulation for the large-diameter lens.

Looking at Fig. 6, it can be seen that when the phase of the S and P-polarization does not differ much, a low error can be achieved. However, if the phase is large in one polarization but small in the other, a large error will occur. This is caused by the fact that the variation in one dimension will affect both polarizations. It will affect the polarization that is parallel with the dimension the most, but it will have a smaller but significant effect on the other polarization. This means that the lookup table is less accurate for large phase differences between polarizations. There are solutions to this problem [6], but this makes fabrication more complex, and even with the larger errors at some meta-atoms, a well enough distinction between polarizations can be made.

D. Small-Scale Full Lens Design

To check if the lookup table works with the final lens design, a simulation in FDTD is done. This is done at a smaller radius than the final design will have, to reduce simulation time and memory used in the simulation. First, the phase profile for the lens is chosen, which is

$$\varphi(x, y) = \frac{2\pi}{\lambda}\left(f - \sqrt{f^2 + x^2 + y^2}\right), \tag{8}$$

where λ is the wavelength of the source, f the focal length of the lens, and x and y the spatial coordinates. This phase profile is identical to that of a spherical lens. This profile is made for both the P and S polarizations, each at its own focal length. This gives two phases for each position in the coordinate system. These phases are then compared to the lookup table, and the meta-atom with the lowest phase error is chosen.

The simulation is done twice, once with the source in the S-polarization and once in the P-polarization. The resulting near and far fields are then checked to see if the correct focusing and focal lengths are achieved.

E. Nearfield Stitching

As mentioned previously, larger lens simulations are very computationally-intensive to run. A workaround for this is to use near-field stitching. This method takes the fields of each meta-atom on its own and stitches them together to get the field of the whole lens.

First, the fields for all meta-atoms are sampled at a lower sampling rate to reduce the complexity of the simulation. After this, the lens is made in the same manner as the full lens simulation, with the fields of each meta-atom placed in the correct position. The fields at the edges of each unit cell are averaged with the fields at the edges of the neighbouring cells.

To check if this leads to the correct phase and amplitude of the fields, a simulation with a radius that is the same as the small-scale full lens simulation is done to compare the two methods. After checking, a larger radius simulation is done. After this, the simulation is extended with a far-field projection to gather intensity profiles at further distances as inputs for the MDPR algorithm.

F. GDS Export

GDS exports are time-consuming for large metalenses due to the large number of elements. To decrease the time required, a method utilizing the polystencil function in ANSYS Lumerical was employed. At a specific height of the meta-atom, this function is used to extract the length and width of that slice of the meta-atom. Since the meta-atoms are uniform over the height, this is all that is necessary to obtain all of the dimensions.

After obtaining the dimensions of all the meta-atoms in the lookup table, the full lens is built from the lookup table in the GDS file.

G. Final Design

The lookup table was made with the previously mentioned constraints, resulting in a working wavelength of $800nm$, a unit-cell size of $350nm$, and a material choice of TiO2 for the meta-atoms and glass for the substrate. The height was set at $1\mu m$ to achieve the largest phase change between the minimal and maximal width and length. A radius of $100\mu m$ was chosen, because smaller radii would make measurement difficult. The focal lengths were set to $1500\mu m$ for the S-polarization and $1300\mu m$ for the P-polarization. The width and length of the meta-atoms vary from $100nm$ to $250nm$. The resulting electric field amplitude over the propagation axis can be seen in Fig. 8 and 9. As can be seen, the focal lengths differ across polarizations. Looking at a slice of the phase profile in Fig. 7, it is evident that it does not align perfectly with the target phase profile. This is caused by the large phase difference between the two polarizations at these places, which can not be achieved. This affects the overall focusing efficiency, but not significantly enough to make the lens unusable for the measurements. This also explains why the peak in Fig. 9 is slightly higher than the one in Fig. 8.

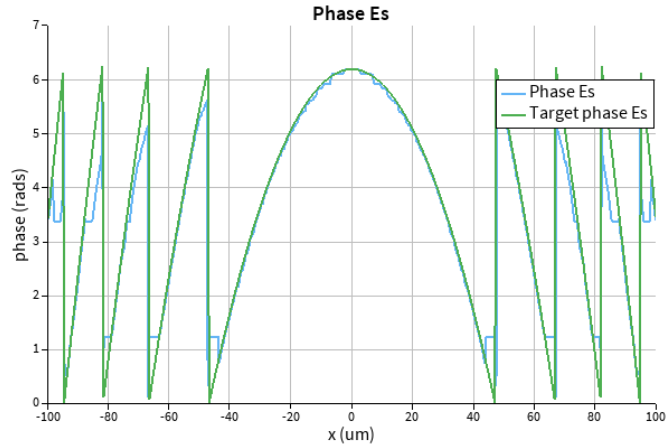


Fig. 7: Slice of the phase profile of the lens obtained from near-field stitching compared with the target phase profile.

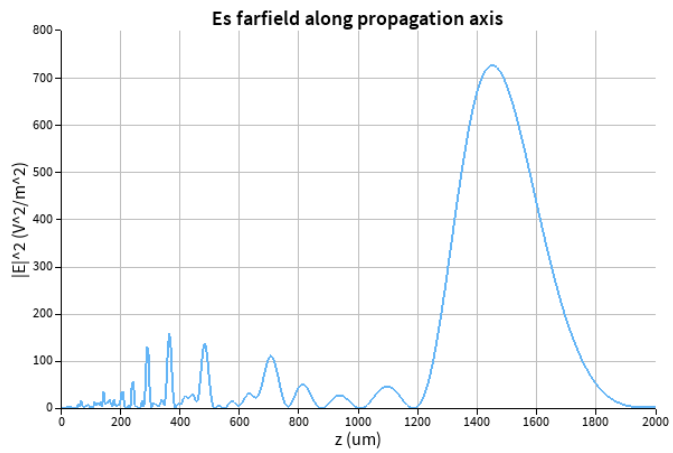


Fig. 8: Intensity of the electric field along the propagation axis when illuminated with S-polarized light.

The phase full profiles for the different polarizations can be seen in Fig. 10 and Fig. 11. Finally, a visualization of the GDS file can be seen in Fig. 12.

IV. Measurement Requirements

The MDPR algorithm has several requirements related to convergence and correct sampling to obtain all high-frequency components of the phase profile.

A. CCD Requirements

In the paper by Liu [12], there are equations derived for the pixel size and the number of pixels of the CCD camera. The pixel size is determined by the highest transverse spatial frequency of the lens, which is given by

$$k_{transverse} = \frac{2\pi}{\lambda}\mathbf{NA}. \tag{9}$$

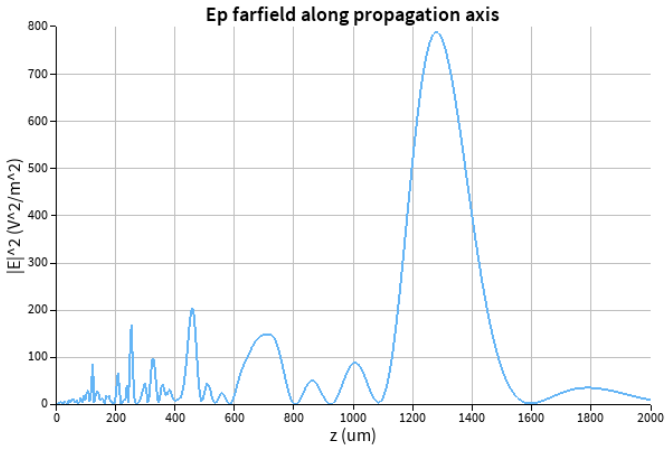


Fig. 9: Intensity of the electric field along the propagation axis when illuminated with P-polarized light.

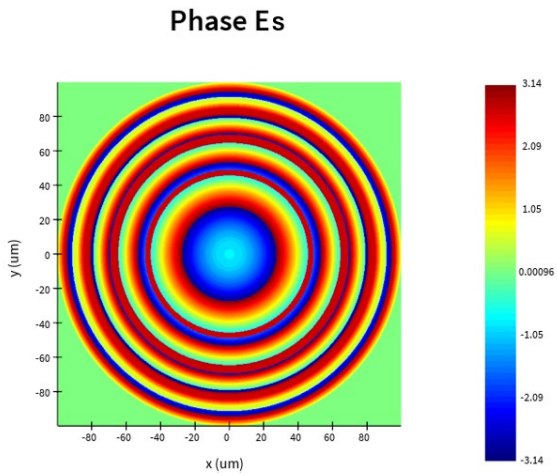


Fig. 10: Phase profile of the S-polarization of the near-field stitched simulation lens with a radius of $100\mu m$.

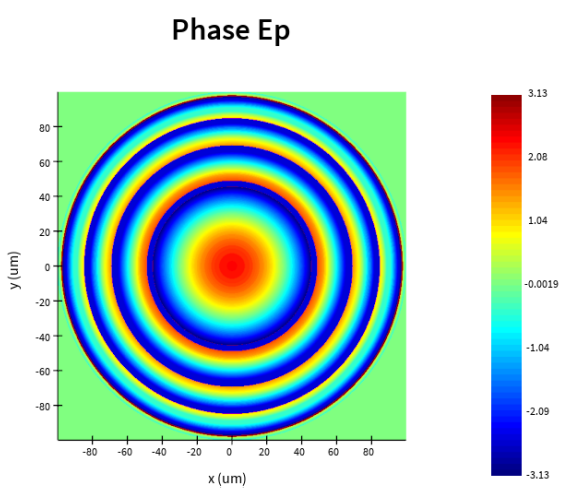


Fig. 11: Phase profile of the P-polarization of the near-field stitched simulation lens with a radius of $100\mu m$.

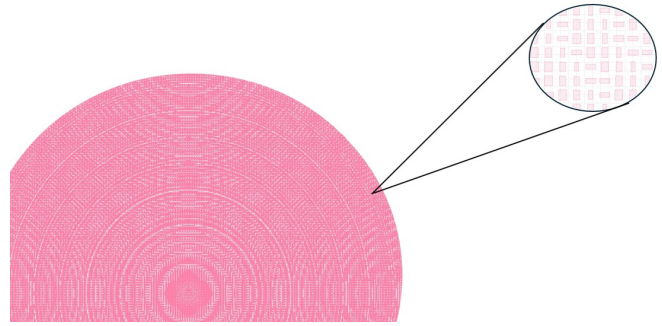


Fig. 12: GDS file visualization.

To satisfy the Nyquist criterion, the sampling frequency should be twice that of the highest frequency in the signal. This gives

$$\frac{2\pi}{a} > 2k_{transverse}, \tag{10}$$

$$\frac{\pi}{a} > \frac{2\pi}{\lambda}\mathbf{NA}, \tag{11}$$

$$a < \frac{\lambda}{2\mathbf{NA}}, \tag{12}$$

where a is the pixel size.

For the number of pixels, the formula for the field of view limitation is used. This formula comes from the need for the CCD to capture the whole intensity profile of the lens. This means that the total area of the CCD must be larger than the area of the metalens. In one dimension, this boils down to

$$N_{pixels} \cdot a > 2r, \tag{13}$$

where N_{pixels} is the number of pixels and r the radius of the metalens. Combining 12 and 13 gives

$$N_{pixels} > \frac{4r\mathbf{NA}}{\lambda}. \tag{14}$$

B. Measurement Distances

There are requirements for the distances at which the intensity measurements are taken for the MDPR to work. This section will be split up into the measurement between two subsequent measurements, d, and the total length of all the measurements combined, z_{total}. Fig. 2 is made for reference.

1) Individual Measurement Distance: For the MDPR algorithm to converge, the distance between measurements needs to be correct. If the distance is too small, there will not be enough change in the intensity profile. This limits the amount of information that is provided to the MDPR algorithm, which will make it unable to converge to the correct phase profile. To make the algorithm converge as quickly as possible, as much distance as possible is needed to provide the most information [16], [17]. The minimal distance is different per sample and is dependent on the focal length of the lens. The maximum distance for each measurement is determined by the z_{total} and $N_{measurements}$ constraints.

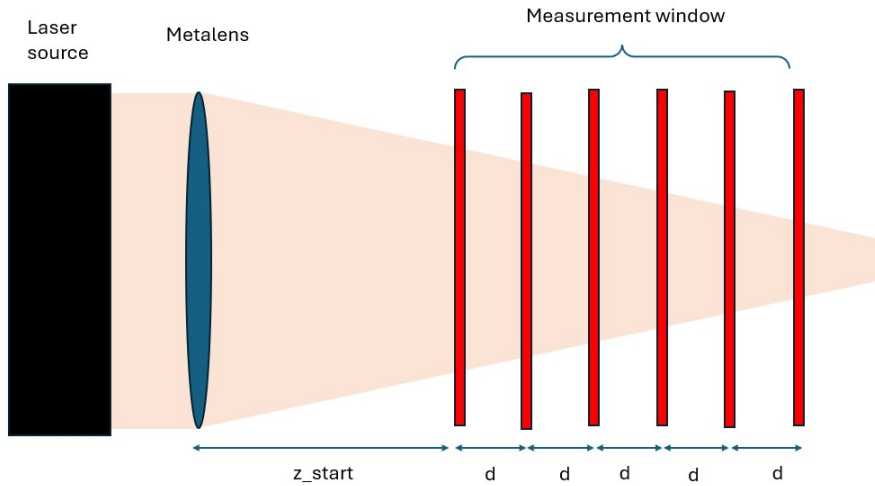


Fig. 13: Visual representation of the second test for z_{max}.

2) Number of Measurements: The algorithm can get stuck in local minima if there are not enough measurements provided. For convergence to the absolute minima, which corresponds to the correct phase profile, [12] reports in the supplementary information that at least 5 measurements are needed to converge correctly. Experimentation with simulated data confirms this, as long as the total scanning length is long enough. Adding more measurements had a positive effect on the accuracy of the algorithm.

3) Total Measurement Distance: The total measurement distance has similar effects on the convergence of the algorithm. Taking it too small will limit the information provided to the algorithm and will prevent it from converging [12]. Therefore, it is best to take it as large as possible. For the characterization of lenses, a good indicator for enough change of the intensity profile to provide enough information is a total measurement length that comes close to the focal length of the lens. However, there is also a constraint on the maximal total distance of the measurement. This is due to the propagation method used in the algorithm, which is the angular spectrum. A detailed analysis of the requirement is given in [18], [19]. The formula comes from the sampling of the transfer function of the propagation kernel used in the angular spectrum method. The spectrum of the transfer function looks like a chirp, where the frequency is dependent on the propagation distance, among other parameters. When the angular spectrum method is used, this kernel is sampled at discrete intervals, and when the sampling does not satisfy the Nyquist criterion, high-frequency information is lost, and the propagation will not be accurate. The formula for satisfying the Nyquist criterion can be rewritten as

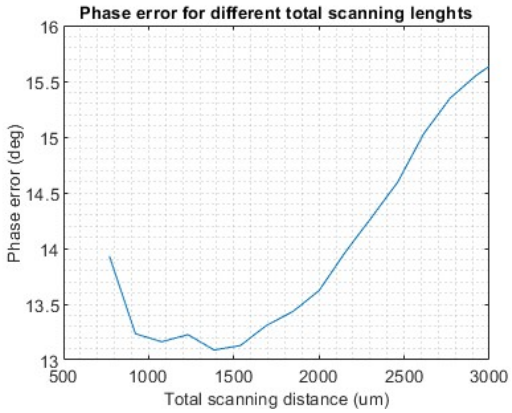
$$z_{total} \leq \frac{N_{pixels} a^2}{\lambda}, \tag{15}$$

To test this constraint of the total scanning distance, a simulation was done that exceeded the total scanning distance given by (15). A metalens with a focal length of $750\mu m$ was simulated, and intensity measurements were done every $75\mu m$. The pixel size was $2\mu m$, the wavelength was $800nm$, and the number of pixels in both directions was 201. This gives a maximum total scanning distance z_{max} for this particular measurement at $1000\mu m$, which is more than the focal length of the lens. This will ensure that the measurements will contain enough information to converge around when the total scanning length is z_{max}.

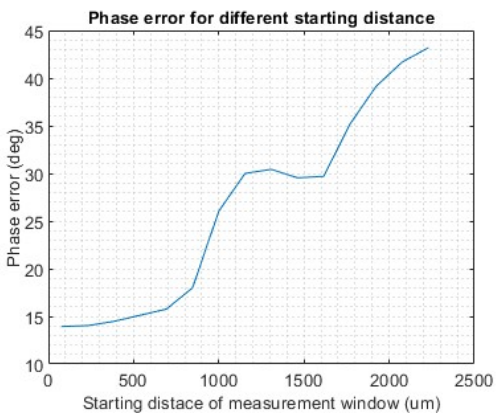
For the first test, the algorithm was fed an increasing number of measurements, increasing the total scanning distance. The phase profile produced by the algorithm is then compared to the phase profile produced by the simulation to test the accuracy. As can be seen from Fig. 14a, increasing the total scanning distance improves the accuracy of the algorithm until around $1500\mu m$. This is expected because adding more intensity measurements adds more data, which makes the algorithm more accurate. After $1500\mu m$, the accuracy gets worse for each increase in the total scanning length.

The second test takes a window of $770\mu m$, which starts at different distances from the sample plane. This ensures that the amount of information is the same for each run of the algorithm; only the starting distance differs. A visual explanation can be seen in Fig. 13. Looking at Fig. 14b, it can be clearly seen that taking a starting distance close to or over z_{max} leads to a large error, which gets worse with increasing starting distances.

Furthermore, the total measurement distance should not exceed twice the focal length. For maximum resolution, the metalens in the sample plane image should span the whole CCD image, meaning that the diameter of the lens should be close to $N_{pixels} \cdot a$. After twice the focal distance, the illuminated spot will be broader than the metalens, and some parts will be cut off from the image, losing this information. Back propagation of this measurement will result in incorrect sample plane reconstruction.



(a) Increasing total scanning distance VS average phase error compared with simulated phase profile.



(b) Increasing starting distance of measurement window of $770\mu m$ VS average phase error compared with simulated phase profile.

Fig. 14: Test results for z_{max} constraint.

V. Results of Algorithm with Simulation

The algorithm was first tested using the large-scale lens simulation from Chapter III. The total measurement distance $z_{total} = 1500\mu m$ and $d = 100\mu m$. The intensity measurements consist of $N_{pixels} = 201$ of size $a = 2\mu m$, resulting in a maximum total scanning distance of $z_{max} = 1654\mu m$. The algorithm was run for 30 iterations. The resulting phase profile for both polarizations and the corresponding absolute errors can be seen in Fig. 15. The average absolute phase error was $6.6°$ for P-polarization and $6.0°$ for S-polarization.

Another simulation was done with a high-NA lens. This lens has a focal length of $135\mu m$. The total measurement distance $z_{total} = 135\mu m$ and $d = 13.5\mu m$. The intensity measurements consist of $N_{pixels} = 501$ of size $a = 0.4\mu m$. The small pixel size was chosen to not exceed the pixel size limit given by (12). The comparison with the phase profile given by the simulation can be seen in Fig. 15c and 15f. The average absolute phase error was $1.65°$ after 40 iterations.

VI. Experiment

To test the algorithm with the lens designed in Chapter III, an experiment was done to capture intensity measurements. The source was a SuperK Fianium FIU-6 PP laser that could produce a range of wavelengths. The CCD was CS135MUN by Thorlabs, which has a pixel size of $a = 4.8\mu m$. The sample was mounted on a DRV250 motorized stage by Thorlabs. A polarization filter, LVPIS100-MP2, was placed just before the sample. Right after the sample was a 20x objective lens to magnify the images, resulting in a pixel size of $a = 0.24\mu m$. Finally, a tube lens is used to image on the CCD. A schematic of the setup can be seen in Fig. 16.

The measurement consisted of 17 measurements with $d = 60\mu m$, resulting in $z_{total} = 1020\mu m$. The wavelength of the laser was set at $650nm$. This was not by choice. The laser failed to produce the desired $800nm$ wavelength, and it was decided to use $650nm$. After these measurements, the polarization filter was turned $90°$, and the experiment was repeated.

The resulting intensity measurements were converted to digital information in MATLAB. The pixel size was digitally enlarged to $2.4\mu m$ to have z_{total} comply with (15).

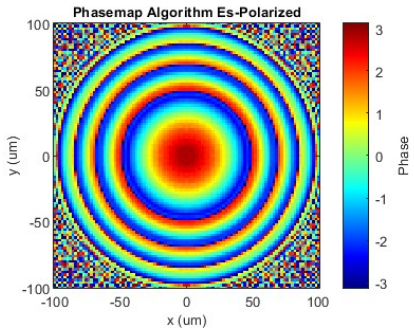
The results of the algorithm with these measurements can be seen in Fig. 17. Some of the measurements were removed using the high noise checking function described in Chapter II. The phase is compared with a perfect phase profile. Because the wavelength of $650nm$ is not the designed wavelength, the focal length is deduced from the intensity measurements. The focal length for the Ep polarized light was $760\mu m$, and for Es polarized light $840\mu m$. The algorithm was run for 200 iterations.

Finally, other measurement data from a similar setup were also used in the algorithm, which can be seen in Fig. 17c. This data was collected with a different experimental setup. The measurements were done at a wavelength of $587nm$. The lens had a diameter of $171.4\mu m$ and a focal length of $10mm$. The algorithm was run for 600 iterations.

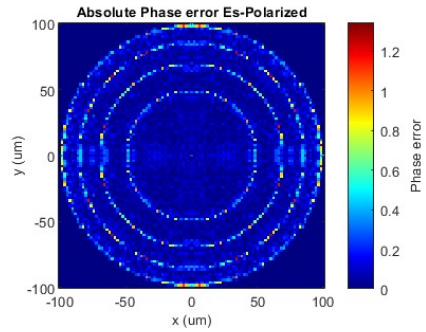
VII. Discussion

The results from the experiment, while showing some resemblance to the expected phase profiles, were worse than expected. There could be multiple causes for this, which will be discussed below. It was not possible to address these causes due to the limited time available to experiment. The metalens fabrication experienced multiple delays, being delivered a couple of days before the deadline of the paper. Also, the laser source broke down during the measurements, forcing the measurements to be done at a different wavelength and reducing the available time further.

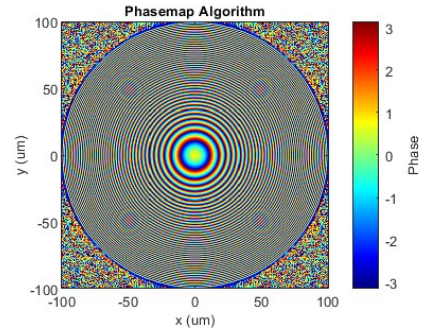
The first reason for the worse-than-expected results is the noise in the measurements. Even with the noise-reducing features of the algorithm, the convergence appeared to get stuck in local minima. The convergence time was significantly larger with the experimental data than with the simulated data. Furthermore, the noise could have also prevented the algorithm



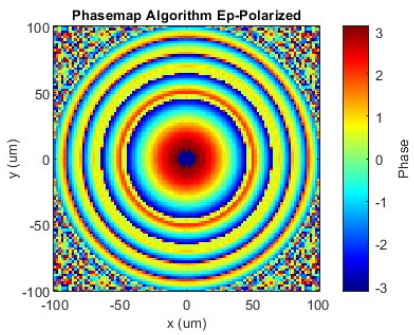
(a) Phase profile produced by algorithm with $z_{total} = 1500\mu m$, $d = 100\mu m$, $a = 2\mu m$ and $N_{pixels} = 201$. The metalens has radius $r = 100\mu m$ and $f = 1500\mu m$ with S-polarized light.



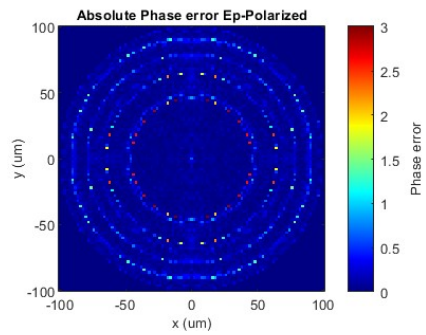
(b) Absolute error when compared with phase profile produced by simulation for the S-polarization.



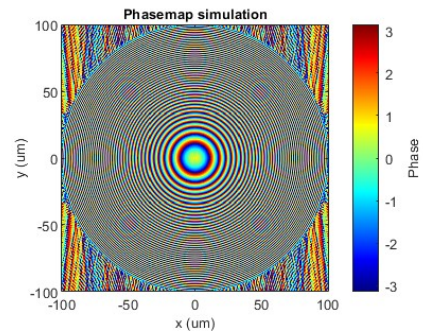
(c) Phase profile produced by algorithm for high-NA lens (**NA** $= 0.6$), with $z_{total} = 135\mu m$, $d = 13.5\mu m$, $a = 0.4\mu m$ and $N_{pixels} = 501$. The metalens has radius $r = 100\mu m$ and $f = 135\mu m$.



(d) Phase profile produced by algorithm with $z_{total} = 1500\mu m$, $d = 100\mu m$, $a = 2\mu m$ and $N_{pixels} = 201$. The metalens has radius $r = 100\mu m$ and $f = 1300\mu m$ with P-polarized light.



(e) Absolute error when compared with phase profile produced by simulation for the P-polarization.



(f) Phase profile produced by simulation for high-NA lens (**NA** $= 0.6$).

Fig. 15: Results from MDPR algorithm with simulated data.

from converging to the right solution in general. More measurements to reduce the noise during the measurements need to be taken to achieve good results with MDPR. In [12], a set of two polarization filters is used to filter out background noise. This is done by putting one filter in front of the sample and one in front of the CCD camera, letting only the right polarization of light reach the camera. The reason that the Es phase profile looks better than the Ep phase profile could be that the gain of the camera was higher during the Ep measurement, increasing the noise.

The second reason is the overexposed pixels. Due to the higher background noise, a higher power from the laser was used to achieve better contrast. This resulted in measurements closer to the focal point of the sample being overexposed. The algorithm can filter these pixels out, but they still result in less information for the algorithm to converge to the proper solution. By lowering the background noise, the power of the laser can be reduced, negating this issue. Alternatively, more measurements could be done to collect more information for the algorithm to use.

The data that was used for Fig. 17c did not have any overexposed pixels in its measurements, and resulted in a phase profile closer to the expected profile. It still was slow to converge, showing that reducing the background noise could improve it even more.

One of the algorithm's features ran into limitations. The high-noise measurement rejection works well and improves convergence for the experimental data. However, for data with overexposed pixels, this feature breaks down. Because the overexposed pixels cannot display the actual intensity of that pixel, propagating those measurements back to the sample plane will always lead to large errors with the other measurements. This makes the user unable to determine whether there is higher-than-average noise present in these measurements.

Finally, the interplay of (12) and (15) makes the pixel size an important parameter to achieve good results with the MDPR algorithm. A way to circumvent this is by zero-padding the data, which increases N_{pixels}. This was done with the simulated data with success, but it still needs to be tested with the experimental data.

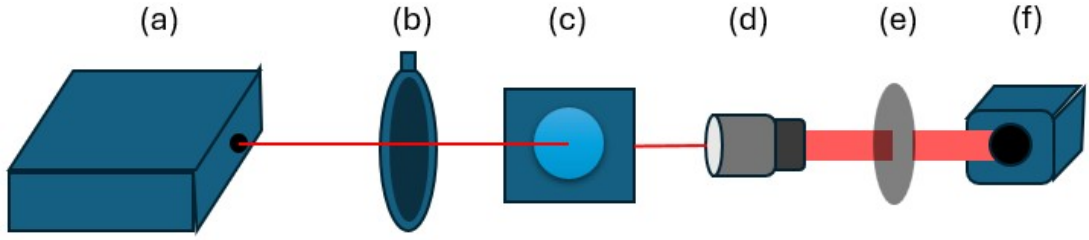
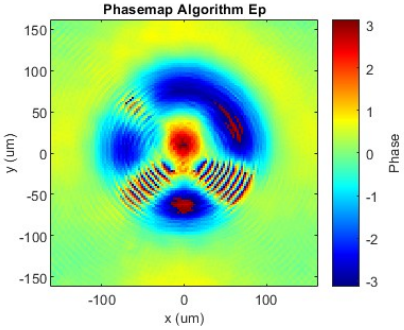
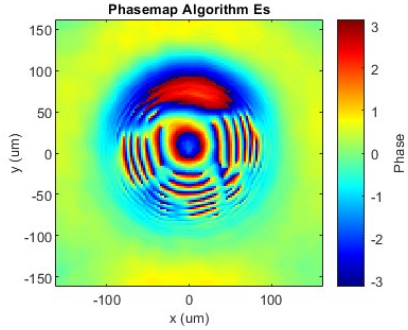


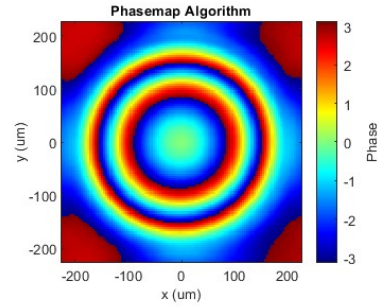
Fig. 16: Schematic of measurement setup. The components visualized are (a) laser source, (b) linear polarizer, (c) metalens sample, (d) objective lens, (e) tube lens, and (f) CCD camera.



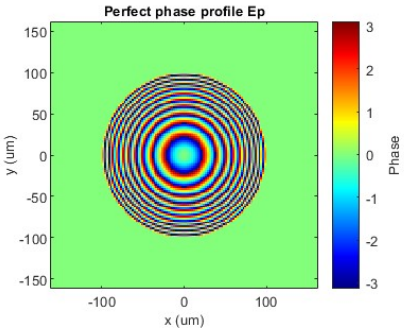
(a) Phase profile produced by the algorithm with experimental measurements for the P-polarization. $z_{total} = 1020\mu m$, $d = 60\mu m$, $a = 2.4\mu m$ and $N_{pixels} = 135$. The metalens has radius $r = 100\mu m$ and $f \approx 760\mu m$ with P-polarized light.



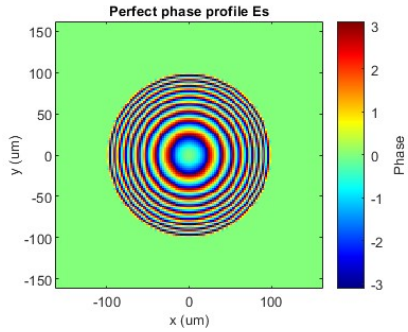
(b) Phase profile produced by the algorithm with experimental measurements for the S-polarization. $z_{total} = 1020\mu m$, $d = 60\mu m$, $a = 2.4\mu m$ and $N_{pixels} = 135$. The metalens has radius $r = 100\mu m$ and $f \approx 840\mu m$ with S-polarized light.



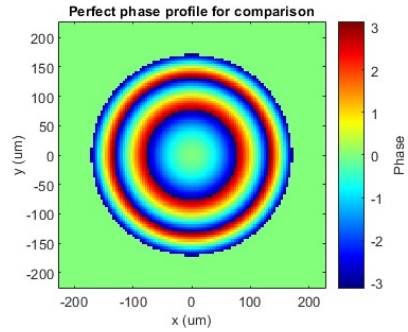
(c) Phase profile produced by the algorithm with experimental measurements from a different setup. $z_{total} = 10.2mm$, $d = 0.6mm$, $a = 4.2\mu m$ and $N_{pixels} = 108$. The metalens has radius $r = 171.4\mu m$ and $f \approx 10mm$.



(d) Perfect phase profile for a focal length of $760\mu m$ for comparison.



(e) Perfect phase profile for a focal length of $840\mu m$ for comparison.



(f) Perfect phase profile for a focal length of $10mm$ for comparison.

Fig. 17: Results from MDPR algorithm with experimental data.

VIII. Conclusion

In this paper, Multi-Distance Phase Retrieval was examined for the characterization of polarization-sensitive metalenses. A method for designing and simulating polarization-sensitive metalenses in ANSYS Lumerical was developed. This method used rectangular meta-atoms to impart different phase profiles to differently polarized light. While the method was verified with small-scale simulations, near-field stitching was ultimately used to simulate and design large-radii lenses, due to its lower computation and memory costs. The final design of the lens had a radius of $100\mu m$, a focal length of $1300\mu m$, and $1500\mu m$ for the different polarization axes, and was designed to work at a wavelength of $800nm$.

The Multi-Distance Phase Retrieval algorithm proposed in [12] was implemented and tested with simulated measurements. Some constraints for the measurements were also examined. The total scanning distance constraint was not yet mentioned in [12], and proved to have a considerable effect on the results. The resulting phase profiles obtained from inputting the data into the algorithm showed an average phase error of $6.6°$ for the S-polarized measurements, and $6.0°$ for the P-polarized measurements when compared with the simulated phase profiles. A high-NA metalens, with an NA of 0.6, was also simulated. The measurements from this resulted in an average phase error of $1.65°$ when compared with the simulated phase profiles. This proves that under simulated conditions, the MDPR algorithm can accurately characterize both polarization-sensitive and high-NA metalenses.

An experiment was done using a measurement setup comprised of a laser source, a polarization filter, an objective lens, a tube lens, and a CCD camera. These measurements were then used as input for the MDPR algorithm. While the results showed some correlation with the expected phase profiles, they did not converge fully. This could be caused by noise, overexposed pixels, and not enough information being carried in the measurements. The data supplied by another experimental setup did not have the problem of overexposed pixels and resulted in a phase profile that was close to the expected one. From this, it can be concluded that the algorithm works, but is sensitive to noise. It also needs enough information to converge, which is more than with the simulated data.

Future work should use two polarizers to suppress background noise, use lower laser power to avoid overexposure of pixels, and increase the number of measurements (or zero-pad the data) to improve convergence.

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